



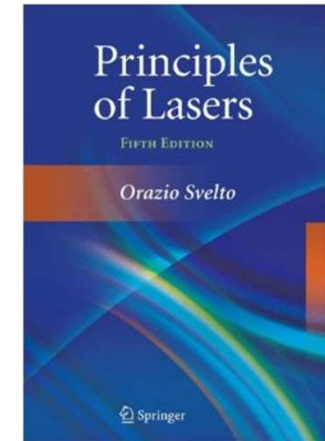
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch6-1

Control & Improvement of Laser Properties



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Chapter 6: Control & Improvement of Laser Performance

❖ 6.1 Mode Selection

- Transverse mode selection
- Longitudinal mode selection

❖ 6.2 Frequency-Stabilized Laser

- External factors which influence the frequency stability
- Frequency stabilization principle: optical length of stable cavity
- Frequency stabilization method

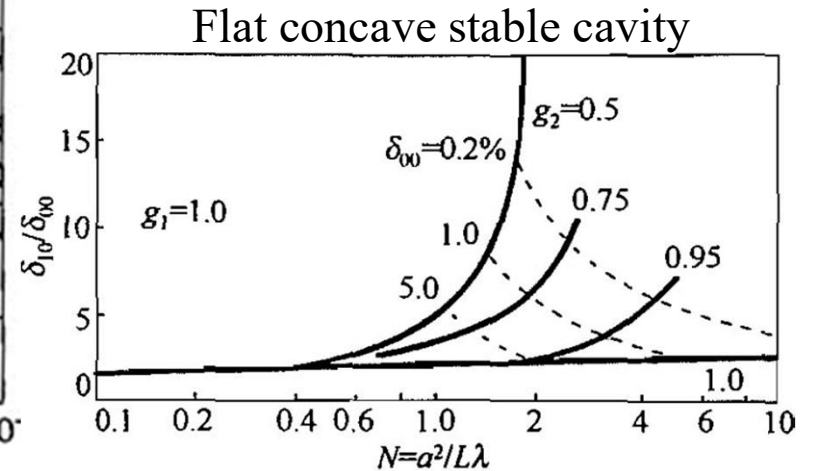
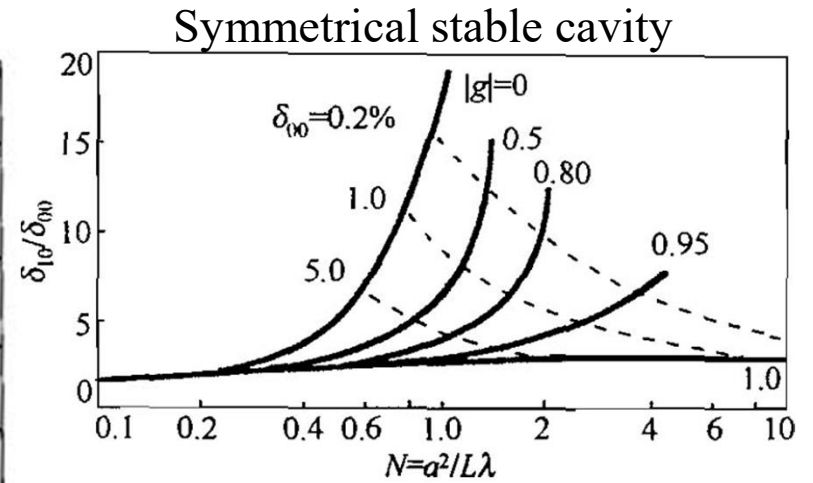
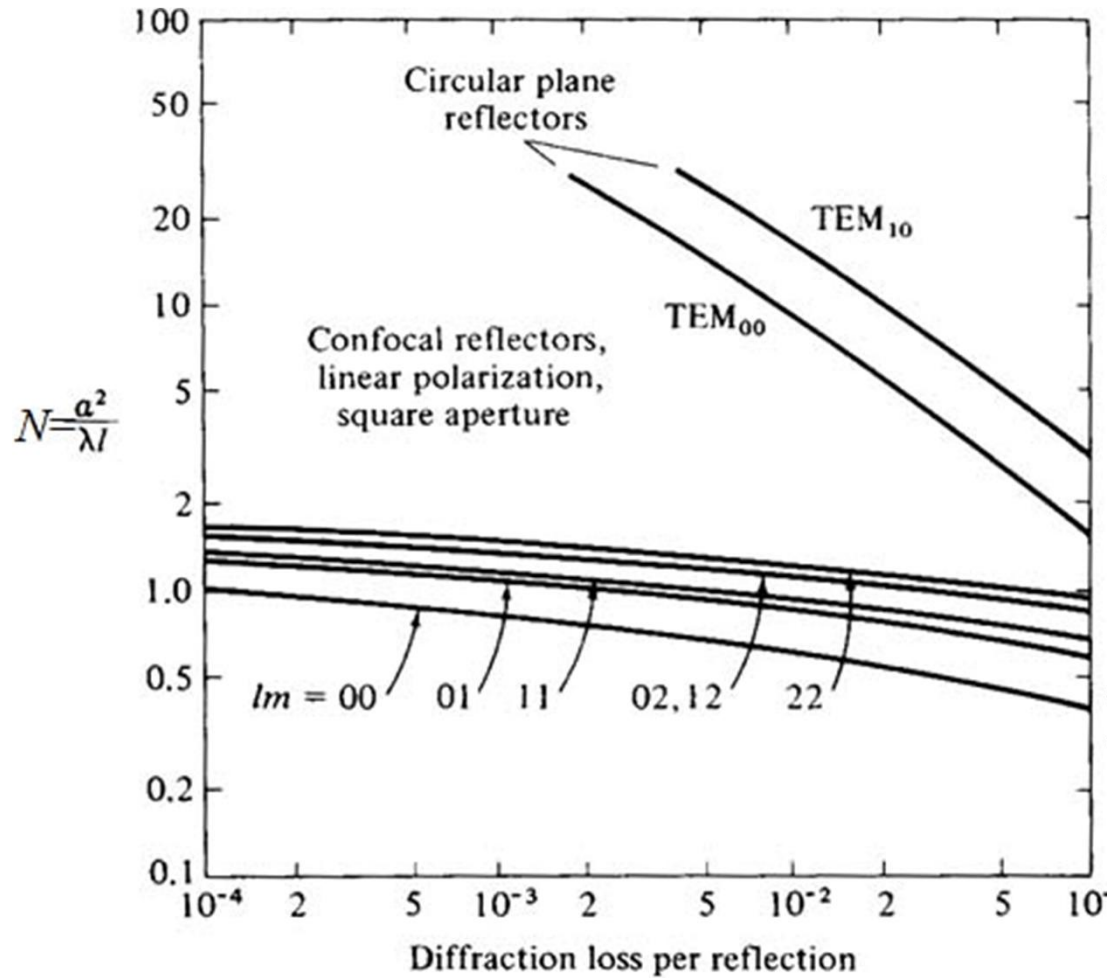
6.1 Mode Selection

Purpose and Motivation: Fundamental transverse mode and single longitudinal mode possess very good spatial and temporal coherence.

❖ Transverse mode selection

- Physics Foundation: different transverse modes have different diffraction losses
- Selection Rules
 - ✓ Maximize the difference in diffraction loss between the fundamental and higher-order modes. Ensure that only the TEM₀₀ mode meets the threshold condition, while suppressing all other modes.
 - ✓ Minimize all losses except for diffraction loss, and maximize the proportion of diffraction loss in the total loss.

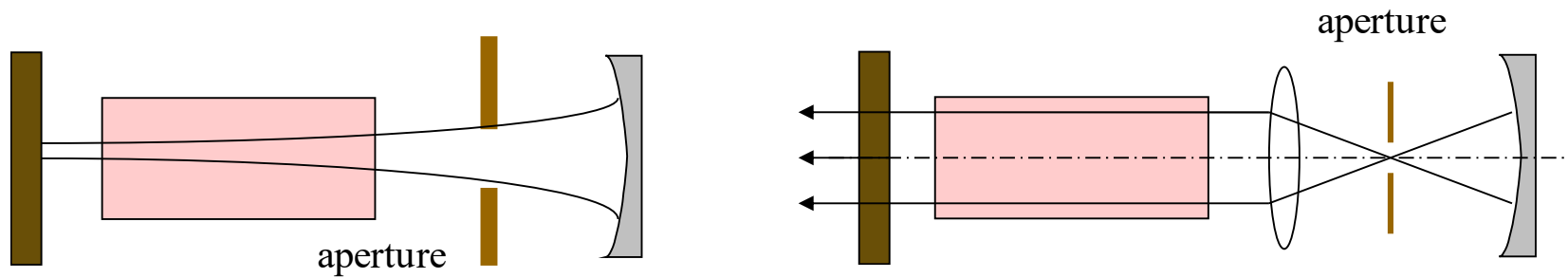
$$\alpha = \alpha_T + \alpha_d + \alpha_{s_1} + \alpha_{s_2} + \alpha_i$$



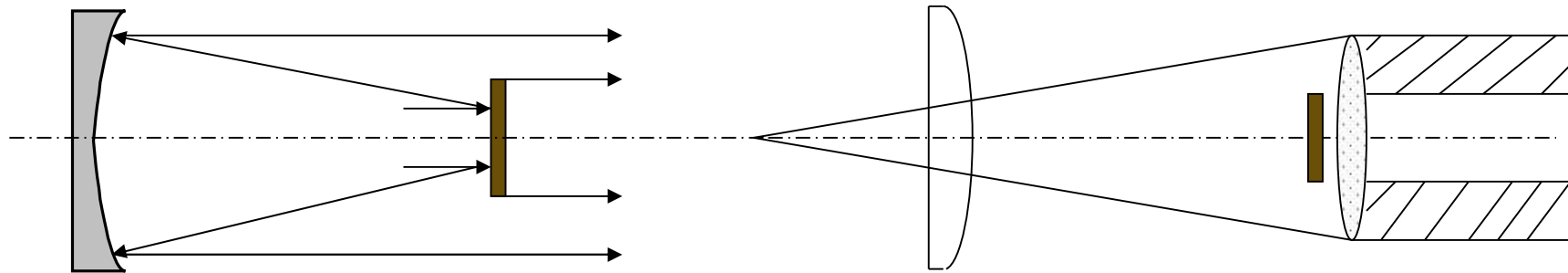
➤ Selection methods for transverse modes

Cavity design; Small aperture; Nonstable cavity; Fine-tuned cavity

Small aperture for transverse mode selection



Nonstable cavity (Suitable for lasers with high gain)



Fine-tuned cavity

Cavity parameter g , N selection

❖ Longitudinal mode selection

- ✓ To achieve single longitudinal mode within the range of specific transition lines

➤ Selection rule

- ✓ Enlarge the gain difference between adjacent longitudinal modes or artificially induce loss difference

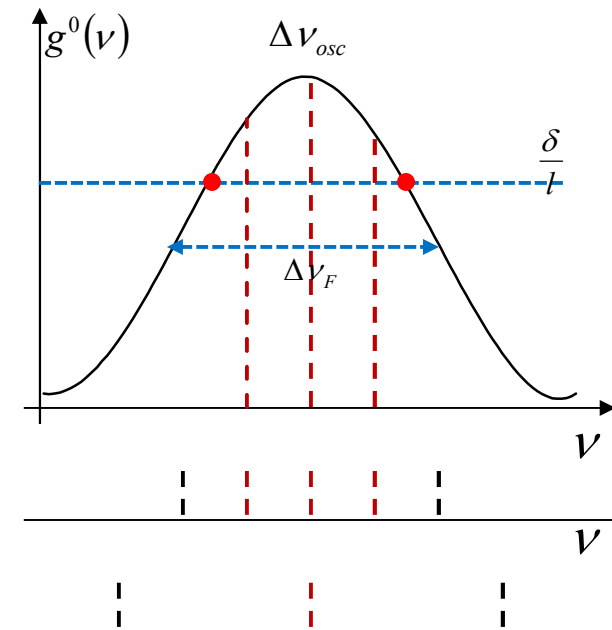
➤ Selection methods for longitudinal modes

- ✓ Short cavity

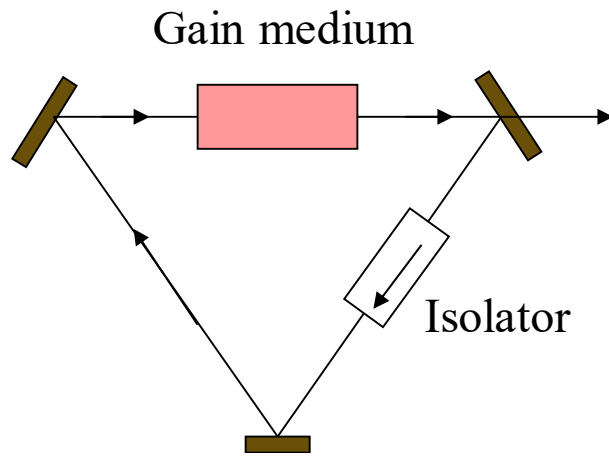
Decrease cavity length, increase the longitudinal mode spacing

$$\Delta \nu_q = \frac{c}{2L'} > \Delta \nu_{osc}$$

Suitable for lasers with narrow fluorescent linewidth, such as He-Ne laser



Travelling-wave cavity



$$\Delta \nu_j = \frac{c}{2\mu d \cos \theta}$$

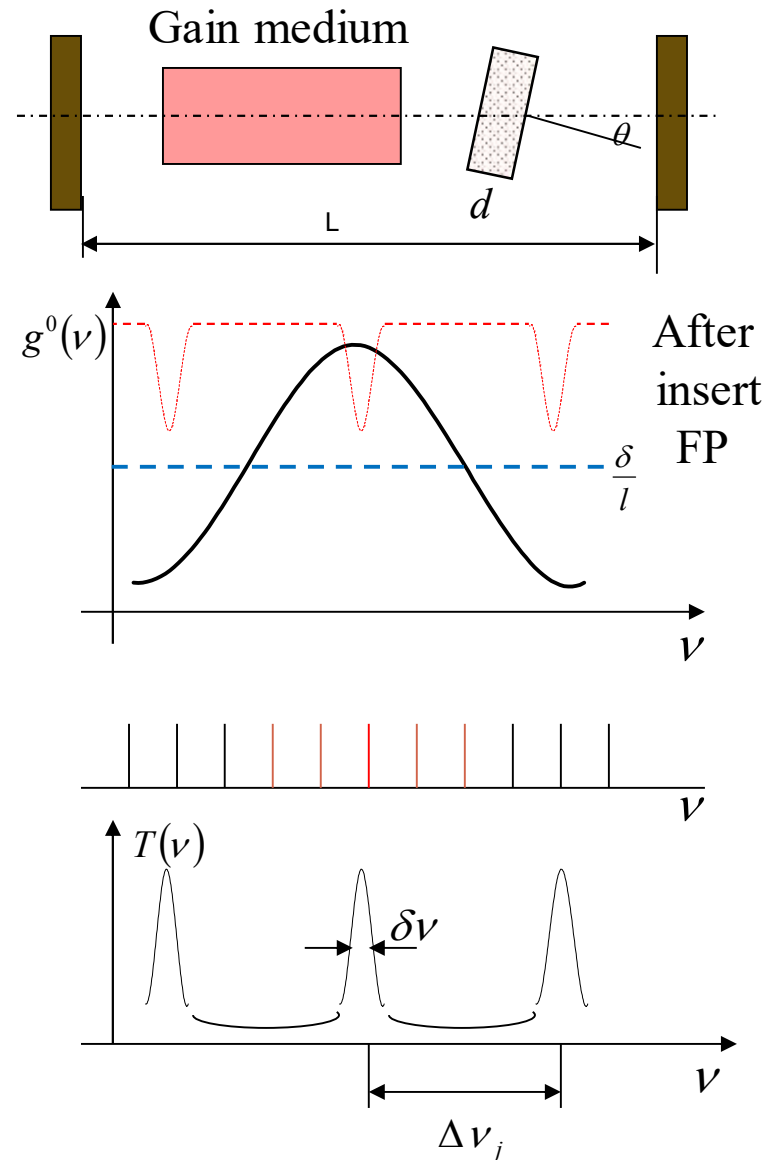
Design requirement
for F-P etalon

$$\delta \nu = \frac{c}{2\pi\mu d} \frac{1-r}{\sqrt{r}}$$

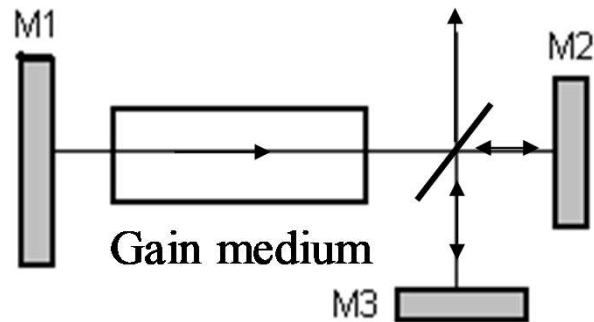
$$\Delta \nu_j > \Delta \nu_{osc}$$

$$\delta \nu < \Delta \nu_q = \frac{c}{2L'}$$

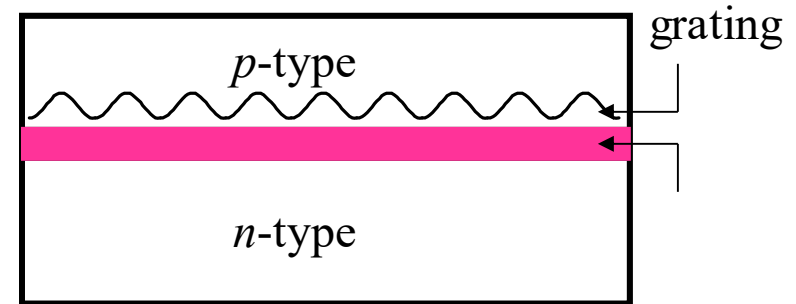
F-P etalon



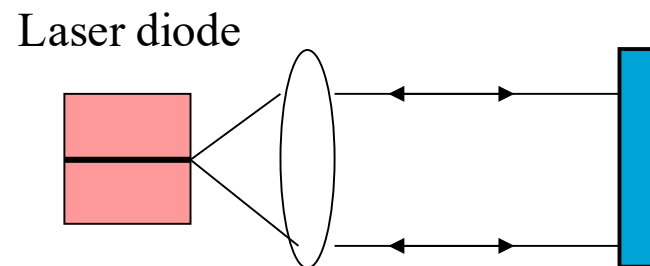
Combined cavity



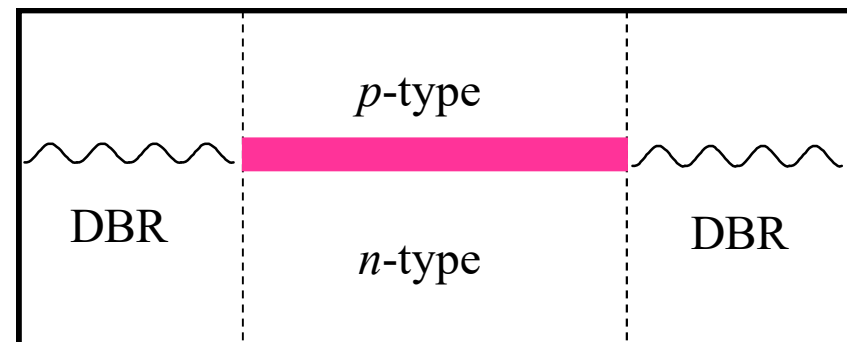
Distributed feedback (DFB) cavity



External cavity laser



Distributed Bragg Reflector (DBR)



6.2 Frequency-Stabilized Laser

❖ External factors which influence the frequency stability

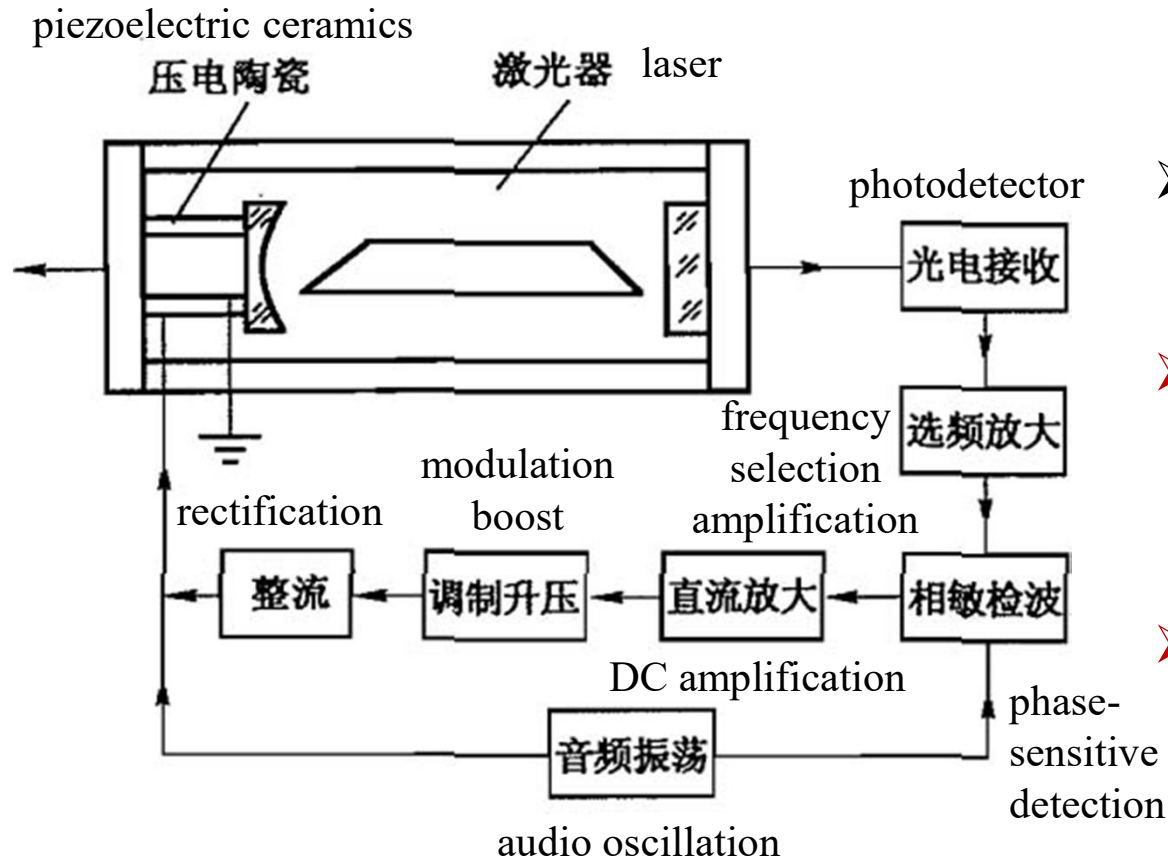
- Longitudinal frequency: $\nu_q = q \frac{c}{2\eta L}$ Frequency stability: $\frac{\Delta\nu}{\bar{\nu}}$
- Variation of cavity geometry: temperature and vibration ($10^{-3}/^{\circ}\text{C}$)
- Refractive index change: temperature, $\Delta\eta$ fluctuation (discharging and driving current, etc.), pressure, humidity
- Frequency stability of single mode He-Ne laser: 10^{-4} - 10^{-5} ($\Delta\nu=10^{10}$ Hz)
- Improved methods include constant temperature, shock proof, sound insulation, voltage stabilization, steady current : 10^{-7}

❖ Frequency stabilization principle: optical length of stable cavity

- Select standard reference frequency
- Error signal acquisition
- Drive electronic servo system to automatically adjust the cavity length

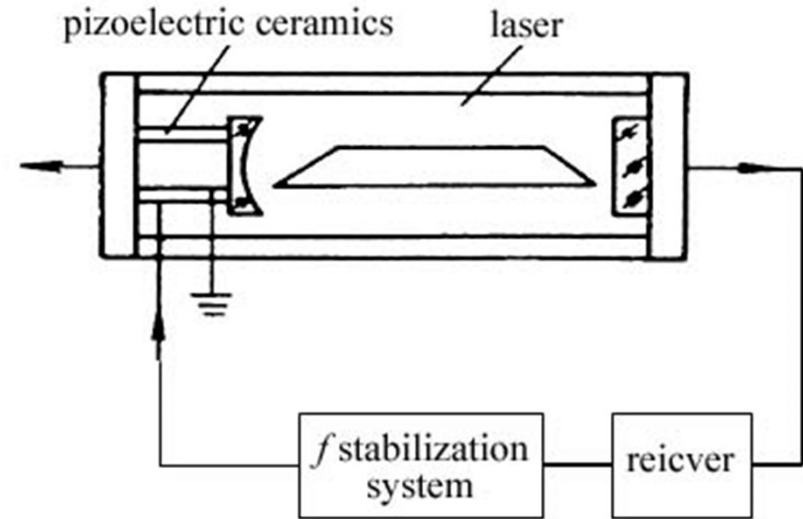
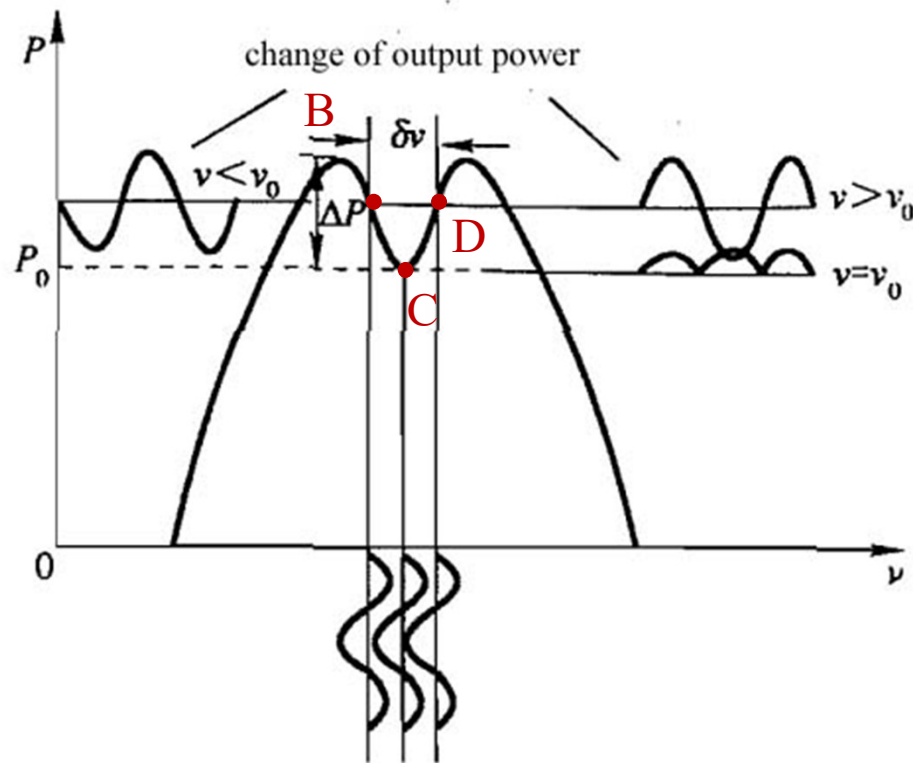
❖ Frequency stabilization method

Lamp dip; Saturated absorption; Zeeman absorption; F-P etalon



- Piezoelectric ceramics: change cavity length
- **Direct-current:** change the single mode laser output frequency
- **AC signal:** signal searching, judge positive or negative

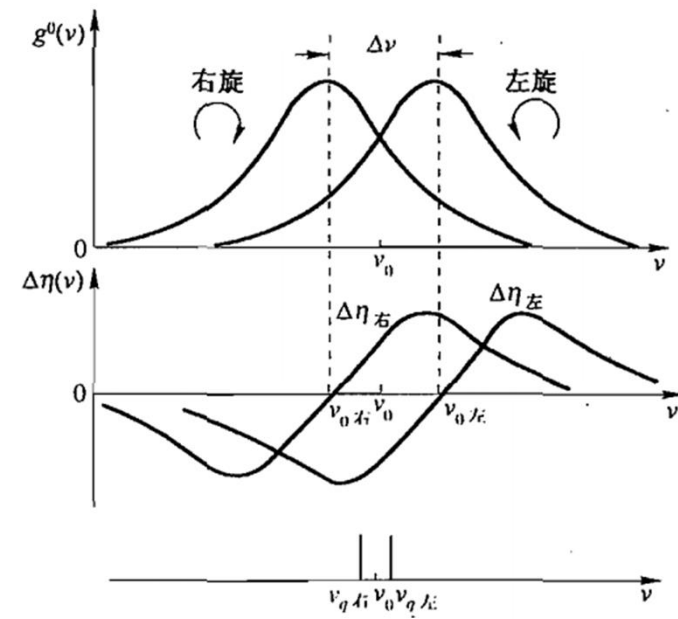
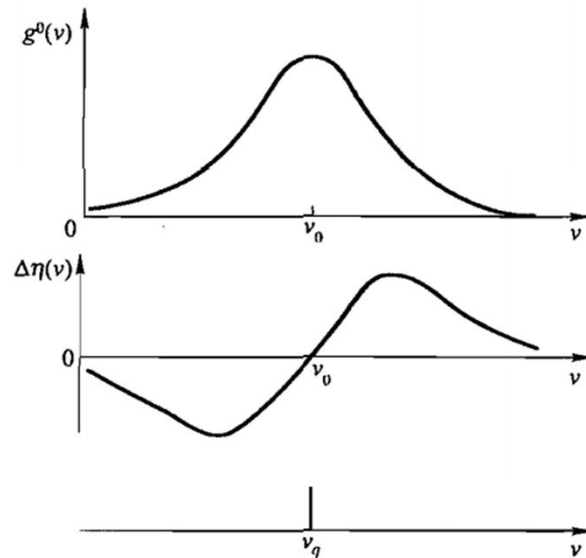
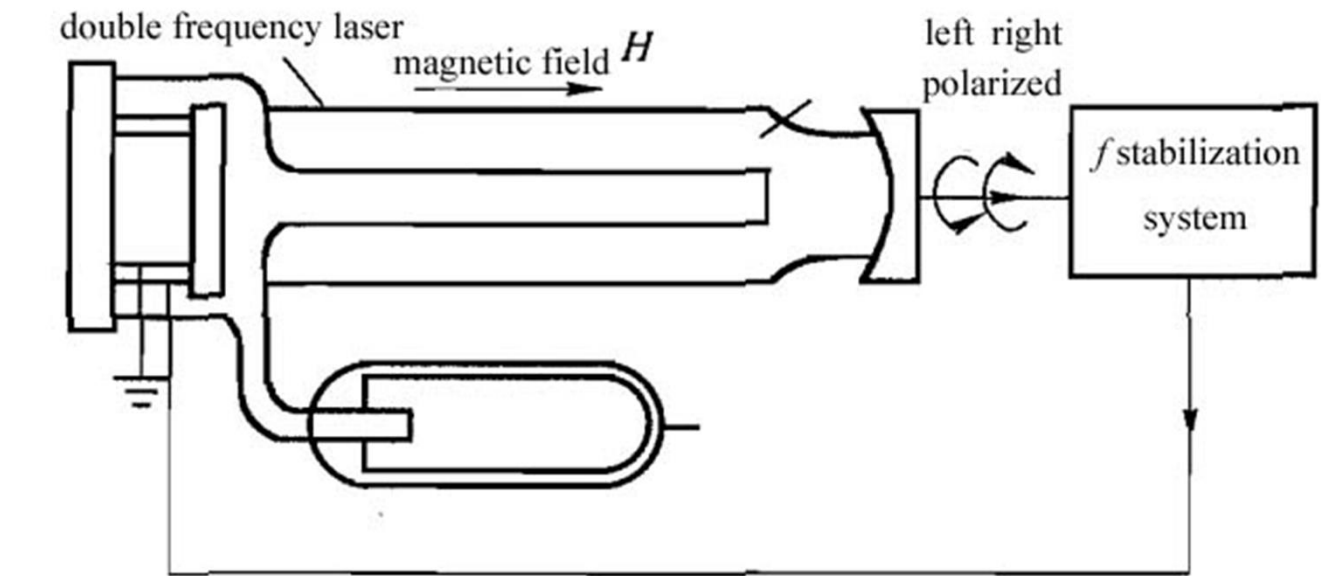
forward voltage, external (+), internal (-), piezoelectric ceramics $\uparrow \rightarrow$ cavity length $\downarrow \rightarrow \nu_q \uparrow$
backward voltage, external (-), internal (+), piezoelectric ceramics $\downarrow \rightarrow$ cavity length $\uparrow \rightarrow \nu_q \downarrow$



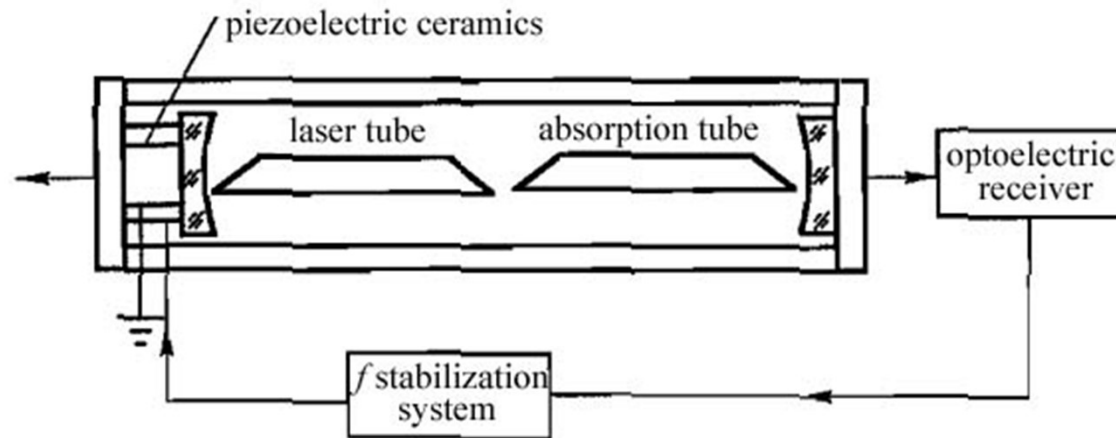
Frequency stabilization: 10^{-9}

- $\nu > \nu_0$ D-point: same phase, reverse V, piezoelectric ceramics $\downarrow \rightarrow$ cavity length $\uparrow \rightarrow \nu_q \downarrow$, pull back ν_0
- $\nu < \nu_0$ B-point: reverse phase, positive V, piezoelectric ceramics $\uparrow \rightarrow$ cavity length $\downarrow \rightarrow \nu_q \uparrow$, pull back ν_0
- $\nu = \nu_0$ C-point: $2f$, 0 voltage, no change of piezoelectric ceramics, $\nu = \nu_0$
- ✓ Requires the symmetry of Lamb dip, narrow and dip (low pressure)
- ✓ Bad frequency reproducibility 10^{-7}

➤ Zeeman frequency stabilization



➤ Saturated absorption-induced frequency stabilization: inverted Lamb dip



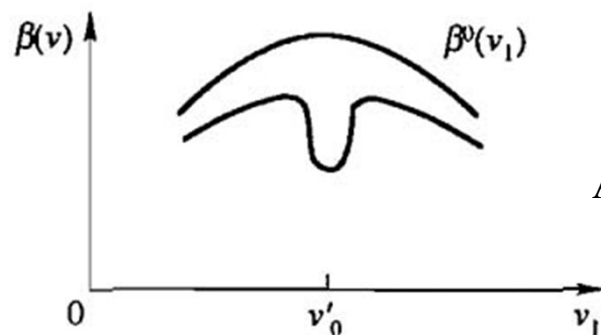
- Absorption tube pressure: 1~10 Pa
- Doppler broadening dominates
- Good frequency stability of absorption peak for low pressure gas

Absorption curve
Hole-burning effect

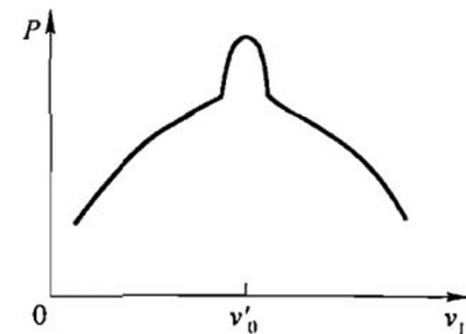
$$\nu_1 = \nu'_0 \quad \beta(\nu_1) = \frac{\beta^0(\nu'_0)}{\sqrt{1 + \frac{I_+ + I_-}{I_s}}}$$

Burning hole width

$$\delta\nu' = \sqrt{1 + \frac{2I_+}{I_s}} \Delta\nu'_H$$



Absorption saturation > gain saturation



Single pass loss of resonant cavity with absorption tube

$$\delta'(\nu_1) = \delta + \beta(\nu_1)L'$$

Output power $P = \frac{1}{2} ATI_s \left[\left(\frac{g_m l}{\delta'} \right)^2 - 1 \right]$



P vs. ν_1 curve forms an inverted lamb dip

- Frequency stabilization: $10^{-11} \sim 10^{-12}$
- Frequency reproducibility: 10^{-11}
- ✓ 632.8 nm: isotopic iodine (碘) vapour
- ✓ 3.39 μm : methane (甲烷)